

AMR MAP EDITOR

USER MANUAL

ROS2 • Fleet Manager • Open-RMF / Traffic Editor

Item	Details
Code package version	v0.11.7 (from package.json)
Source reviewed	AMR-Map-Editor-Path-Direction-v0.6.0 ZIP
Manual scope	All user-facing functions found in the supplied ZIP
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1. Application Overview and Workflow

AMR Map Editor is a browser-based CAD/GIS-style editor for ROS occupancy maps, AMR navigation topology, operational zones, robot configuration, and Open-RMF / Traffic Editor building geometry.

Stage	User action
Import	Load ROS YAML + PGM/PNG/JPG/JPEG or image-only map
Edit Map	Paint or erase occupancy raster
Objects	Place navigation objects and set heading
Paths	Draw lanes, connect junctions, set direction/orientation/Graph Index
Zones	Create operational zones including Keepout
Building	Create Wall, Door, Floor, Model and Measurement
Robot	Configure footprint / clearance / turning radius / speed
Preview	Review complete project overview
Validate	Run geometry/topology checks
Save / Database	Save locally, export .amrmap, or store .amrmap + RMF Bundle through database API
Export	Generate ROS, JSON, GeoJSON, RMF, nav_graph and bundle formats

1.1 Recommended Workflow

1. Import the map and verify metadata.
2. Edit occupancy pixels if required.
3. Create navigation objects.
4. Create paths and connect real topology junctions.
5. Set path type, travel direction, orientation, Graph Index, speed, width and clearance.
6. Create zones; use Keepout where a keepout mask is required.
7. Create RMF building geometry.
8. Configure the robot.
9. Preview and validate.
10. Save the editable project.
11. Export the deployment formats or save a database revision.

2. Project Lifecycle and Toolbar

The top toolbar exposes the main project and editor commands.

Command	Behavior
New Map	Creates a new untitled project; if dirty, prompts Save / Don't Save / Cancel
Open .amrmap / JSON	Loads a project file through schema migration/validation
Import Map	Imports occupancy map files
Save	Saves current project to browser IndexedDB
Save As .amrmap	Downloads a complete editable project file
Undo / Redo	Restores editor project snapshots
Select / Pan / Brush / Eraser / Path / Zone / Measure	Primary canvas tools
Zoom In / Out / Fit Map	Viewport controls
Validate	Runs project validator
Preview	Opens map overview and entity detail list
Help	Opens embedded PDF manual
Database	Opens PostgreSQL API project manager
Export	Opens export dialog
Deploy	Currently blocked; no deployment API target is configured

2.1 Dirty / Save State

The header displays Draft / Unsaved and local save status. The application also performs delayed automatic local saving after edits.

2.2 How to use

1. Use New Map when starting a clean project.
2. Use Import Map for ROS occupancy files or image-only maps.
3. Choose the canvas tool from the toolbar before editing.
4. Use Preview and Validate before saving or exporting.
5. Use Save for local browser persistence, Save As .amrmap for a portable editable file, Database for server revisions, and Export for deployment files.

3. Importing Maps

Supported map selections are YAML/YML plus occupancy image, or an image-only map.

Input	Behavior
YAML/YML + image	Reads ROS metadata and matches the referenced image among selected files
PGM	Parses P2/P5-style occupancy data through the PGM parser and retains semantic occupancy
PNG/JPG/JPEG	Loads raster dimensions and derives occupancy representation
Image without YAML	Prompts for Resolution, Origin X, Origin Y and Origin Yaw before preview

3.1 Import Preview

Before applying the map, the dialog shows source file, image, pixel size, resolution, world size, and origin. If the current project already has objects/paths/zones, the user can either replace the map and keep those entities or replace the map and clear them.

3.2 How to use

1. Select the ROS YAML and its image together, or select an image by itself.
2. For image-only import, enter resolution and origin values.
3. Review image dimensions, map size and origin in the preview.
4. Choose whether existing objects/paths/zones should be kept or cleared.
5. Confirm Import, then verify coordinates on the canvas before creating navigation data.

Important

Incorrect resolution or origin shifts world coordinates, RMF geometry, reference coordinates, nav graphs and keepout masks.

4. Occupancy Map Editing

The editor modifies the occupancy raster directly in the browser while preserving semantic occupancy information.

Tool / setting	Implemented behavior
Brush shapes	Freehand, Line, Rectangle, Polygon
Brush sizes	1, 3, 5, 10, 20, 50 px
Brush color	User-selectable RGB color; obstacle semantics remain occupied even when color is custom
Eraser	Restores original map pixels in the erased region
Polygon brush	Click points; Enter, double-click or Apply Polygon to finish; Clear cancels current polygon
Line / Rectangle	Drag start to end / opposite corner

4.1 Semantic Occupancy

Obstacle brush writes occupied semantics, while erasing recovers semantic occupancy from restored original pixels. This keeps validation and PGM generation aligned with the edited map.

4.2 How to use

1. Choose Brush or Eraser from the toolbar.
2. Select brush shape, size and color.
3. For Freehand, drag directly on the map. For Line or Rectangle, drag from start to end.
4. For Polygon, click each corner and finish with Enter, double-click, or Apply Polygon.
5. Use Eraser over edited areas to restore original imported pixels.

5. Navigation Objects and Heading

- Waypoint
- Home
- Charging Station
- Docking Station
- Pickup
- Drop-off
- Waiting
- Parking

Property	Behavior
Name / ID / Type	Identity fields
X / Y	World coordinates in meters
Heading / Yaw	Heading editor stores yaw in radians; display supports degree/heading labels
Description	Free text
Enabled	Controls active/exported state where applicable
Pickup	RMF pickup_dispenser metadata
Drop-off	RMF dropoff_ingestor metadata
Duplicate / Delete	Inspector actions

5.1 Placement

Choose an object type in the Objects library, then click the map. Object labels and heading orientation are visible on the canvas and in Preview.

5.2 How to use

1. Choose the desired navigation object type.
2. Click the map to place it.
3. Select the object and edit name, coordinates, heading/yaw, description and enabled state.
4. For Pickup/Drop-off, set the RMF dispenser/ingestor property when required.
5. Use the heading arrow to confirm robot facing direction.

6. Paths, Direction, Orientation and Connectivity

Path types implemented in the editor: Normal, Preferred, One-way, Bidirectional and Restricted.

Property / action	Behavior
Path Type	normal / preferred / one_way / bidirectional / restricted
Lane Orientation	None / Forward / Backward
Graph Index	Non-negative integer used for nav_graphs/N.yaml and .building.yaml graph_idx
Max Speed	m/s
Path Width	meters
Safety Clearance	meters
Points	Individual path vertices are selectable/editable; Delete/Backspace removes a selected point when path remains valid
Reverse Direction / Vertices	One-way shows Reverse Direction; other paths reverse vertex order
Delete	Removes path

6.1 Travel Direction vs Orientation

Travel direction controls allowed A->B / A<->B traversal. Orientation controls robot heading constraint while traversing. They are separate settings.

6.2 Connectivity and Snapping

- Path draw snapping to navigation objects, existing path vertices and path segments.
- Default path snap distance is 0.25 m.
- Dropping onto the middle of an existing segment can insert a junction.
- Connected endpoints are canonicalized to shared coordinates.
- Connect Start, Connect End and Connect Both are available in the Path Connectivity card.
- A compatible path can be merged with the nearest path.
- Green CONNECTED / snap indicators represent actual topology, not only visual overlap.

6.3 RMF Lane Vertex Deduplication

RMF exporters reuse a shared lane vertex when connected endpoints are within approximately 0.02 m, avoiding visually duplicated Traffic Editor junctions caused by floating-point differences.

6.4 How to use

1. Choose Path and click points to draw a lane; finish the path.
2. Select the path and set path type, max speed, width and safety clearance.
3. Set One-way/Bidirectional for permitted travel and Forward/Backward for robot facing direction.
4. Use Connect Start, Connect End or Connect Both to create real topology connections.
5. Drag endpoints onto existing vertices/segments to snap and create junctions.
6. Set Graph Index for multi-nav-graph projects and use Reverse when stored A-to-B order must be flipped.

7. Zones and Keepout

Zone type	Purpose / behavior
No-Go	Hard restricted area; validator rejects paths entering it
Keepout	Hard restricted area plus keepout mask export
Slow	Operational speed zone
Restricted	Restricted operational area
Parking	Parking area
Loading	Loading area
Unloading	Unloading area
Human Traffic	Human-traffic area
Safety	Safety area

7.1 Zone Properties

Zones support Name, Type, Max Speed, vertex count, Duplicate and Delete.

7.2 Keepout Validation

The validator treats enabled Keepout zones like No-Go zones for path intersection. Conflicts are also warned when No-Go/Keepout overlaps Slow, Parking, Loading or Unloading zones.

7.3 Keepout Mask Export

Enabled Keepout polygons are rasterized black on a white map-sized PNG. A companion YAML uses trinary mode, map resolution/origin, negate=0, occupied_thresh=0.65 and free_thresh=0.196.

7.4 How to use

1. Choose Zone and select the required zone type.
2. Click polygon vertices around the operational area and finish the polygon.
3. Select the zone to edit name, type, enabled state and max speed where applicable.
4. Use Keepout when the area must be included in map_keepout.png.
5. Run Validate to check path intersections and zone conflicts.

8. Measure Tool

The canvas has two temporary measurement modes:

Mode	Behavior
Distance	Click multiple points; shows cumulative polyline distance in meters
Area	Click polygon points; shows polygon area in m ²
Clear	Clears the current measurement

The Measure tool is a temporary canvas utility and is separate from RMF Building Measurement entities.

8.1 How to use

1. Select Measure from the toolbar.
2. Click points on the map to create a measurement.
3. Read distance values while adding points.
4. Finish/close the shape when area measurement is needed.
5. Press Escape or clear the draft to exit without modifying navigation/building geometry.

9. RMF Building Geometry

Tool	Creation / main properties
Wall	Click 2 points; Name, Start/End X/Y, Length, Angle, Resize Anchor, Texture, Height, Width, Texture Scale, Alpha, Enabled
Door	Click 2 points; snaps to walls; Door Type, Motion Axis, Motion Degrees, Direction, Plugin, Right/Left Ratio, Enabled
Floor Area	Click 3+ points; finish with Enter/double-click; Texture, Ceiling Texture, Scale, Rotation, Indoor, Enabled
Model	Click once; Model Name, X/Y/Z, Yaw, Static, Dispensable, Enabled
Measurement	Click 2 points; stores start/end and measured distance

9.1 Wall Editing

Selected walls expose draggable START and END handles. Dragging updates geometry live; drag end can snap to another wall endpoint. Numeric Length/Angle editing supports Start, Center and End resize anchors.

9.2 Door-to-Wall Snapping

Door endpoints snap to wall endpoints/segments. After the first door point anchors to a wall, the next point is constrained to that wall with a larger snap range so the door remains collinear.

9.3 Building Configuration

Building configuration includes Building Name, Reference Level, Level and Elevation. The largest enabled valid Floor is used automatically as reference geometry for reference-coordinate generation.

9.4 How to use

1. Choose Wall, Door, Floor, Model or Measurement from Building tools.
2. Draw the entity on the map.
3. Select it and edit Properties.
4. For Walls, change length/angle numerically or drag START/END handles.
5. For Doors, snap the endpoints to the intended wall and verify alignment.
6. For Floor, create a valid enabled polygon when automatic reference coordinates are required.

10. Layers, Selection and Snapping

Layer	Visibility / lock support
Occupancy Map	Yes
Grid	Yes
Building Geometry	Yes
Paths	Yes
Waypoints	Yes
Stations	Yes
Zones	Yes
Robot Footprint	Yes
Validation	Yes
Labels	Yes

10.1 Snap Indicators

- Path SNAP indicator for object/path targets.
- Wall SNAP indicator for wall endpoints.
- DOOR SNAP indicator for wall endpoint/segment targets.

Layer locking prevents interactive editing for the corresponding entity layer while visibility can be toggled independently.

10.2 How to use

1. Use Layers to show/hide or lock entity categories.
2. Use Select to pick and edit entities.
3. Drag endpoints near valid targets until the snap indicator appears.
4. Verify path endpoints show CONNECTED when topology must be shared.
5. Do not rely on visual overlap alone.

11. Robot Configuration

The current project stores one active robot configuration in the editor.

Property	Unit / role
Profile name	Robot configuration name
Width / Length	meters
Footprint	rectangle / circle / polygon model type in project schema
Safety Margin	meters
Inflation Radius	meters
Minimum Clearance	meters
Minimum Turning Radius	meters
Max Speed	m/s

Path clearance validation compares path width against robot width + 2 x safety margin.

11.1 How to use

1. Open Robot Configuration.
2. Enter robot footprint/width values.
3. Set safety clearance, minimum clearance, turning radius and maximum speed.
4. Run Validate and review path clearance warnings.
5. Adjust robot or path parameters before deployment.

12. Preview Mode

Preview renders a read-only project overview fitted to the dialog and includes building geometry, zones, paths, objects and labels.

12.1 Summary Counts

- Objects
- Paths
- Zones
- Walls
- Doors
- Floors
- Models
- Measurements

12.2 Detail Lists

The side panel lists building entities with coordinates/properties, navigation objects with X/Y + heading/yaw, paths with direction/point count/max speed, and zones with type/vertex count/max speed.

12.3 How to use

1. Click Preview.
2. Inspect overall map placement and summary counts.
3. Review object/path/zone/building details in the side panel.
4. Close Preview and correct any issue before validation or export.

13. Validation

Validation returns Error / Warning / Info issues with optional object ID, position and suggested fix.

Area	Checks implemented
Objects	Duplicate IDs, outside map, inside occupancy obstacle
Paths	Duplicate IDs, <2 points, zero-length segment, obstacle intersection, No-Go/Keepout intersection, path width vs robot clearance, disconnected endpoint, invalid one-way direction
Zones	Duplicate IDs, <3 vertices, self-intersection, outside map
Zone conflicts	No-Go/Keepout overlapping Slow/Parking/Loading/Unloading

Validation uses semantic occupancy when available. When it is not available, the code can fall back to seeded factory obstacle rectangles.

13.1 How to use

1. Click Validate after major changes and before export.
2. Review errors first, then warnings and informational messages.
3. Locate the referenced entity/geometry and correct it.
4. Re-run Validate until deployment-blocking issues are resolved.

14. Save, Open, Autosave and Undo/Redo

Editable project state is stored in the .amrmap project format and can also be persisted locally in IndexedDB.

Function	Behavior
Local Save	Stores AMRMapProjectFile in IndexedDB database amr-map-editor / store projects
Autosave	Dirty projects trigger delayed local save after approximately 1.5 seconds
Open	Reads .amrmap / JSON and runs project migration/schema validation
Save As	Downloads .amrmap
Before unload	Browser warns when project has unsaved changes
Undo history	Up to recent project snapshots; commit is used before editing actions
Redo	Restores redo snapshot
New Project	Resets map/entities/building data and creates a new UUID

14.3 How to use

1. Use Save to store the project in browser IndexedDB.
2. Use Save As .amrmap to download a portable editable file.
3. Use Open .amrmap / JSON to restore a saved project.
4. Use Undo/Redo for recent editor changes.
5. Use Copy/Paste/Duplicate on supported selected entities.

15. Export Formats

The Export dialog accepts a base filename and adds the correct suffix/extension automatically.

Output	Scope / notes
AMR Project	Complete .amrmap project
ROS Map (.zip)	ROS YAML + occupancy map package
Occupancy PNG	Current occupancy raster
PGM	Generated ROS occupancy PGM
YAML	ROS map metadata referencing <base>.pgm
RMF Building (.building.yaml)	Traffic Editor/Open-RMF building
RMF Nav Graphs (.zip)	nav_graphs/N.yaml for all used Graph Index values
RMF Bundle (.zip)	Deployment bundle; see Section 19
Reference Coordinates (.yaml)	4 paired RMF/robot corners
GeoJSON (.geojson)	Graph/coordinate exchange
Waypoints JSON	Navigation objects
Paths JSON	Navigation paths
Zones JSON	Zones
Navigation JSON	Map metadata + objects + paths + zones + robot config

15.1 Export Selected

Export Selected is available for Waypoints JSON, Paths JSON, Zones JSON, Navigation JSON and GeoJSON when there is an active selection.

15.2 How to use

1. Open Export from the toolbar.
2. Enter the desired outer export base filename.
3. Choose the required format.
4. For selected-only JSON/GeoJSON exports, select entities before opening Export.
5. For RMF formats, verify Building, Paths, Graph Index and map metadata first.
6. Download the generated file and inspect its contents before deployment.

16. RMF .building.yaml Export

The building exporter targets Traffic Editor style `reference_image` coordinates and converts world-meter entities using resolution, ROS origin/origin yaw and image Y-axis conventions.

16.1 Main Data

- Building Name / Level / Reference Level / Elevation
- Drawing filename
- Floors, walls, doors, models and measurements
- Navigation vertices and lanes
- Pickup/Drop-off RMF parameters
- Crowd-sim scaffold / navmesh filename metadata

16.2 Lane Parameters

```
bidirectional: [4, true/false]
graph_idx: [2, N]
orientation: [1, forward/backward/empty]
speed_limit: [3, value]
```

Graph Index in the Path inspector is written into `graph_idx` and is also used to choose `nav_graphs/N.yaml`.

16.3 How to use

1. Set Building Name, Level, Reference Level and Elevation.
2. Create required building geometry and navigation paths.
3. Set Graph Index, direction, orientation and speed on paths.
4. Open Export and choose RMF Building (.building.yaml).
5. Verify the exported file in Traffic Editor/Open-RMF tooling.

17. Reference Coordinates

Reference coordinates are generated automatically from building Floor geometry; manual coordinate entry is not required.

1. Find enabled Floor polygons with at least 4 vertices.
2. Select the largest valid Floor.
3. Derive exactly 4 principal outer corners.
4. Use the same four physical source points for RMF and robot/world coordinates.
5. Write pairs in Top-left, Top-right, Bottom-right, Bottom-left order.

17.1 Units / conversion

robot[] uses world coordinates in meters. rmf[] uses reference-image conversion scaled into meters.

```
reference = worldToReferenceImage(worldPoint)
rmf_x_m = reference_x_px * resolution
rmf_y_m = -reference_y_px * resolution
```

17.2 Output Style

The exporter intentionally writes compact YAML coordinate pairs as - [x, y]. It rejects output unless there are exactly 4 RMF points and 4 robot points.

17.3 How to use

1. Create and enable the main Floor polygon.
2. Ensure it has at least four unique outer vertices.
3. Verify map resolution/origin and Building Name.
4. Choose Reference Coordinates (.yaml) in Export.
5. Confirm exactly four rmf and four robot coordinate pairs.

18. RMF Nav Graph Export and Multiple Graphs

The browser directly generates RMF nav graph YAML from current NavigationPath and NavigationObject data.

18.1 Graph Generation

1. Group enabled paths by Graph Index.
2. Convert each path point to RMF nav coordinates.
3. Deduplicate vertices within approximately 0.02 m.
4. Convert segments into directed lanes.
5. Add orientation_constraint when Forward/Backward is configured.
6. Add speed_limit when positive.
7. If a lane segment crosses an enabled RMF door, add door_name.
8. Attach navigation-object properties to matching graph vertices.

18.2 One-way / Bidirectional

One-way creates one directed lane. Bidirectional creates forward and reverse directed lanes. Reverse orientation constraints are flipped appropriately.

Graph Index	Output
0	nav_graphs/0.yaml
1	nav_graphs/1.yaml
2	nav_graphs/2.yaml
Unused index	No empty YAML is generated

18.3 Nav Graph ZIP

The RMF Nav Graphs ZIP contains nav_graphs/N.yaml for used indices and a README describing generated graph indices and Graph Index relation to .building.yaml.

18.4 How to use

1. Create and connect enabled paths.
2. Assign Graph Index to every path that belongs to a specific nav graph.
3. Set direction, orientation and speed as required.
4. Open Export and choose RMF Nav Graphs (.zip).
5. Extract the ZIP and verify only the expected N.yaml files exist.
6. Check that bidirectional paths generate two directed lanes and one-way paths generate only the permitted direction.

19. RMF Bundle and Keepout Mask Files

The current bundle implementation uses a fixed internal base name map. The downloaded outer ZIP name still follows the user-entered export base name.

19.1 Exact Bundle Contents

```
map.building.yaml  
map.png  
map.pgm  
map.yaml  
map_keepout.png  
map_keepout.yaml  
nav_graphs/0.yaml  
nav_graphs/1.yaml ...
```

19.2 Bundle Preconditions

- Valid occupancy map image
- Positive resolution and valid map dimensions
- At least one enabled path with 2 or more points, so at least one nav graph exists

19.3 Keepout Files

map_keepout.png is white free space with enabled Keepout polygons painted black. map_keepout.yaml references that image and uses trinary ROS map settings. These files are generated by the current RMF Bundle even when there are no Keepout zones (the mask remains all white).

19.4 How to use

1. Verify the occupancy map is valid and resolution is positive.
2. Ensure at least one enabled path with 2 or more points exists.
3. Create Keepout zones if a keepout mask is required.
4. Set Graph Index values and validate the project.
5. Open Export and choose RMF Bundle (.zip).
6. Extract the ZIP and verify map.building.yaml, map.png, map.pgm, map.yaml, map_keepout.png, map_keepout.yaml and nav_graphs/.

20. Project Database Integration

The frontend contains a Project Database dialog and API client. The backend implementation is external to this supplied ZIP, but the expected API behavior is explicit in the frontend source.

Action	Frontend behavior
Health / Refresh	GET /api/health, then GET /api/projects
Save as New	Generates current .amrmap + RMF Bundle in memory and POSTs multipart form to /api/projects
Save Revision	Generates new .amrmap + RMF Bundle and POSTs to /api/projects/:id/files
Load	GETs latest amrmap, imports it through project importer, then loads it into editor
Bundle	GETs latest rmf_bundle and downloads it
Delete	DELETE /api/projects/:id

20.1 Stored Revision Metadata Expected by UI

The UI expects per-file summary fields: fileType, revision, fileName, sizeBytes, sha256 and createdAt. Project summary includes id, name, description, createdAt, updatedAt and files.

20.2 API Base URL

The API base is read from VITE_API_BASE_URL; if missing, the current code falls back to <http://localhost:3001/api>.

20.3 How to use

1. Open Database and confirm Connected.
2. Enter a project name and use Save as New for the first revision.
3. After edits, use Save Revision to create a new stored revision.
4. Use Load to restore the latest .amrmap.
5. Use Bundle to download the latest RMF Bundle.
6. Use Delete only when the database project should be removed.

21. Help / Embedded User Manual

The Help dialog imports the PDF as a Vite asset and displays it inline using an viewer. It also provides Open in new tab and a fallback link when inline PDF preview is unsupported.

The current source still labels the embedded manual as v0.11.0; after replacing the PDF asset with this updated manual, update HelpDialog text if you want the visible version label to match.

21.1 How to use

1. Click Help.
2. Read the manual in the inline viewer.
3. Use Open in new tab for a larger view.
4. Use the fallback document link if inline preview is unavailable.

22. Keyboard Shortcuts

Shortcut	Action
V	Select
H	Pan
B	Brush
E	Eraser
P	Path
Z	Zone
M	Measure
Ctrl/Cmd + S	Save locally to IndexedDB
Ctrl/Cmd + Z	Undo
Ctrl/Cmd + Shift + Z	Redo
Ctrl/Cmd + D	Duplicate selected object/path/zone
Ctrl/Cmd + C / V	Copy / Paste selected object/path/zone
Enter	Finish path/zone, brush polygon, or building Floor depending active tool
Delete / Backspace	Delete selected path point when one is selected
Delete	Delete selected entity/entities
Escape	Cancel current drawing/measurement/brush draft and return to Select

23. Current Deployment Button Behavior

The toolbar contains a Deploy button, but the supplied source explicitly reports:

Deployment blocked: no deployment API target is configured.

Therefore the editor currently supports file/database preparation for deployment, but direct deployment to a robot/fleet endpoint is not implemented in this ZIP.

23.1 How to use

Use Export or Database to prepare deployment artifacts. Do not rely on Deploy until a deployment API target is implemented and configured.

24. Deployment / Export Checklist

- ☐ Map resolution, origin X/Y and origin yaw are correct.
- ☐ Occupancy raster and semantic obstacle data are correct.
- ☐ Navigation objects are placed correctly and required RMF object names are set.
- ☐ Path endpoints that must connect are true topology connections, not visual overlaps.
- ☐ Path direction and Forward/Backward orientation are correct.
- ☐ Graph Index is correct for every path.
- ☐ Path width and safety clearance are compatible with robot configuration.
- ☐ No-Go and Keepout geometry is correct.
- ☐ Building Name / Level / Reference Level / Elevation are correct.
- ☐ Walls / Doors / Floors / Models / Measurements are correct.
- ☐ Reference Floor is enabled and valid if Reference Coordinates are required.
- ☐ Validation has no unresolved deployment-blocking errors.
- ☐ A newly exported .building.yaml has no unintended duplicate lane vertices.
- ☐ Expected nav_graphs/N.yaml files are present.
- ☐ RMF Bundle contains map.building.yaml, map.png, map.pgm, map.yaml, map_keepout.png, map_keepout.yaml and nav_graphs/.
- ☐ Database Save/Load/Bundle is tested if database workflow is used.
- ☐ Target RMF / robot deployment workflow has been tested with newly exported artifacts.